

MODEL EVALUATION + MODEL OPTIMIZATION

20. Model Evaluation - The process of measuring how well a trained machine learning model performs on unseen data.

Train Model



Predict



Compare predicted with actual



Evaluation metrics

Evaluation metrics depend on whether the problem is:

Regression / Classification

21. Confusion Matrix - A table comparing actual classes with predicted classes.

		Predicted	
		+ve	-ve
Actual	+ve	TP	FN
	-ve	FP	TN

TP (True Positive) - Correctly predicted positive

TN (True Negative) - Correctly predicted negative

FP (False Positive) - Predicted +ve but actually negative

FN (False Negative) - Predicted -ve but actually positive

22. Classification Metrics

i. Accuracy: Percentage of all predictions that are correct.

$$= \frac{TP + TN}{TP + TN + FN + FP}$$

Best when classes are reasonably balanced.

ii. Precision: Among predicted positives, how many are actually positive.

$$= \frac{TP}{TP + FP}$$

Useful when False positives are costly.

iii. Recall: Among actual positives, how many the model correctly identified.

$$= \frac{TP}{(TP + FN)}$$

Useful when False Negatives are costly.

→ Sensitivity: Another name for Recall.

$$= \frac{TP}{TP + FN}$$

iv. Specificity: Percentage of actual negatives correctly identified.

$$= \frac{TN}{TN + FP} \quad [0.63]$$

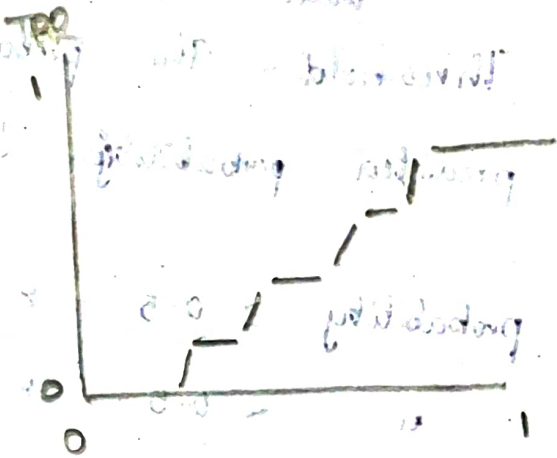
n. F1 Score : Harmonic mean of precision & recall

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Useful when classes are imbalanced.

25. ROC curve and AUC

ROC curve : Shows how True Positive Rate and False Positive Rate change across different classification thresholds.



True +ve Rate

$$TPR = \frac{TP}{TP + FN}$$

same as recall

False +ve Rate

$$FPR = \frac{FP}{FP + TN}$$

ROC - AUC : Measures the classifier's overall ability to rank positive observations above negative ones across thresholds.

→ AUC near 1 : - strong separation

→ AUC near 0.5 - roughly random ranking

24. Precision - Recall Curve - Shows the relationship b/w precision and recall at diff. classification thresholds.
→ especially for: Highly imbalanced datasets.

PR-AUC: Summarizes performance across the

Precision - Recall curve

→ For strongly imbalanced classification:

PR-AUC can be more informative than ROC-AUC.

25. Classification Threshold - The probability cutoff used to convert predicted probabilities into classes.

Typical:

probability ≥ 0.5 → positive

" < 0.5 → negative

26. Regression Metrics -

MAE (Mean Absolute Error): Average absolute prediction error.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

actual predicted

less sensitive to large errors than MSE.

MSE (Mean Squared Error): Average squared prediction error.

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

large errors receive a stronger penalty.

RMSE (Root Mean Squared Error): Square root of MSE.

↓
has same units as target.

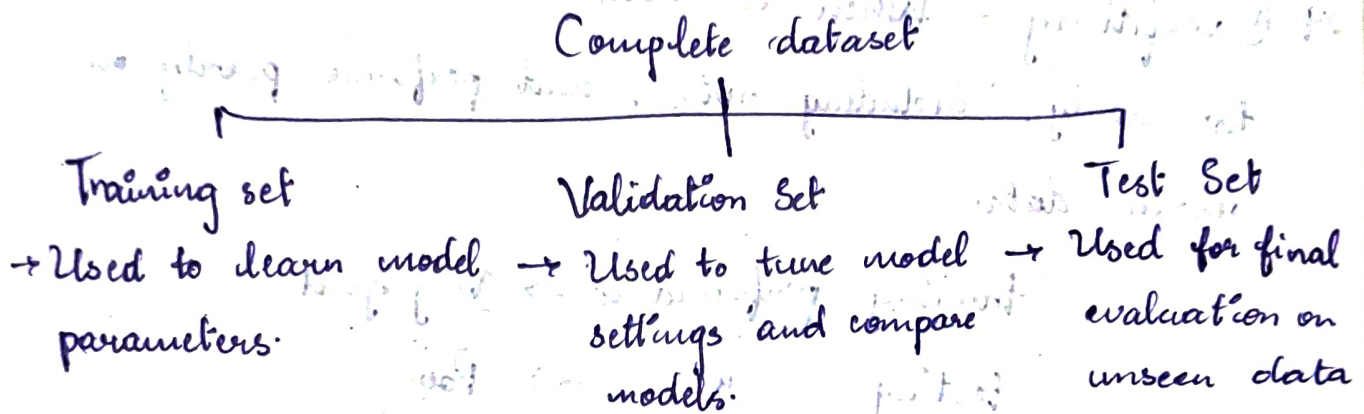
$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

iv. R-Squared : Measures the proportion of target variation explained by a regression model

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

→ Adjusted R^2 : penalizes adding unnecessary predictor variables

27. Train, Validation and Test Data



28. Cross-Validation : A model evaluation technique that repeatedly trains and validates a model on different subsets of the dataset.

Purpose : • more reliable estimate of model performance

⇒ K-fold Cross-Validation ↓

The dataset is divided into k groups.

Example : 5-Fold Cross Validation

Fold 1 → Validation

Fold 2

" 3 → Training

" 4

" 5

Then rotate

→ Final performance is usually the average across all folds.

⇒ Stratified K-Fold

that preserves approximately the same class proportions in each fold.

Useful for:

Classification especially imbalanced classes.

29. **Overfitting** - When a model learns the training data too closely, including noise, and performs poorly on unseen data.

training performance → Very good
testing " → Poor

→ Possible causes:

→ Solutions:

- Model too complex
- Too many features
- Too little training data
- Training too long
- Regularization
- More data
- Feature selection
- Cross-validation
- Reduce model complexity
- Early stopping for neural networks.

30. **Underfitting** - When a model is too simple to capture important patterns in the data.

Training Performance → Poor
Testing " → Poor

→ Solutions:

- Increase model complexity
- Add useful features
- Reduce excessive regularization
- Train appropriately.

31. Bias and Variance

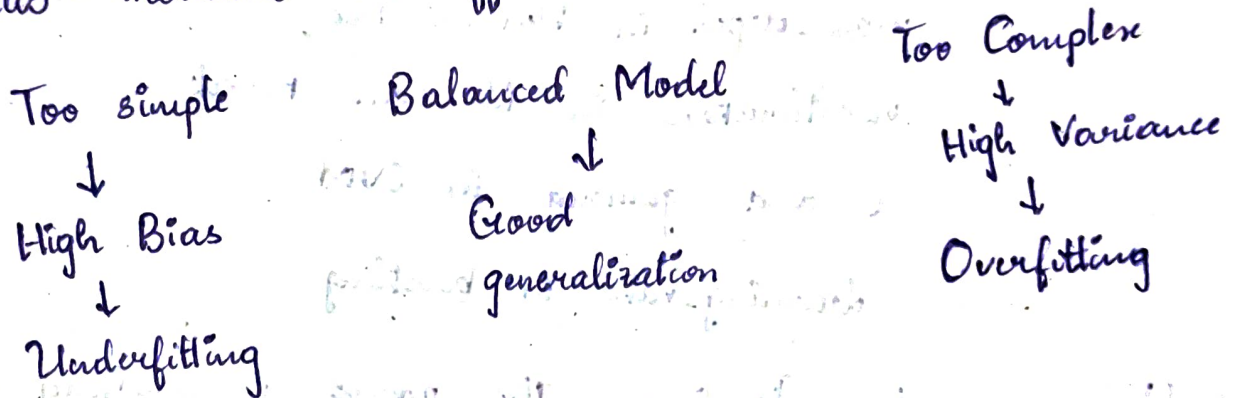
* Bias - Error caused when a model is too simple and makes overly strong assumptions.

High Bias $\rightarrow$ Underfitting

* Variance - Error caused when a model is too sensitive to the training data.

High Variance $\rightarrow$ Overfitting

* Bias - Variance Tradeoff



Goal : Balance Bias + Variance

32. Generalization - The ability of a model to perform well on new unseen data.

** The real goal of ML is :

$\rightarrow$ Not memorize training data

$\rightarrow$ But generalize to new data

33. Regularization - A technique that reduces overfitting by penalizing overly large model parameters.

Main types :

- L1 $\rightarrow$ Lasso
- L2 $\rightarrow$ Ridge

L1 + L2 $\rightarrow$ Elastic Net

L1 Regularization - can make some coefficients exactly 0.

$$\text{Cost} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^m |w_j|$$

L2 Regularization - shrinks coefficients toward zero.

$$\text{Cost} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^m (w_j)^2$$

34. Hyperparameters - Model settings chosen before or during training rather than learned directly from the data.

Examples: K in KNN

max_depth in Decision Tree

n_estimators in Random Forest

C and gamma in SVM

learning_rate in boosting.

35. Hyperparameter Tuning - The process of searching for hyperparameter values that produce better model performance.

Main methods:

- Grid Search

- Random Search

→ Grid Search - Tests every specified combination of hyperparameter values

$$C = [0.1, 1, 10]$$

$$\text{gamma} = [0.01, 0.1, 1]$$

↓

try combinations

↓

select best

Advantage: Systematic

Limitation: Can become computationally expensive.

→ Random Search - Tests randomly selected hyperparameter combinations

Advantage: Usually faster for large search spaces.

36. Data Leakage - Occurs when information unavailable during real-world prediction accidentally influences model training.

Example:

Full dataset
↓
Scale everything
↓
Train/Test Split

X leakage

Better:

Split Dataset
↓
Fit preprocessing ONLY
on training data
↓

Transform validation/test data

37. Model Selection - The process of choosing the model that generalizes best to unseen data

→ Do not simply choose: Highest training accuracy X

→ Instead consider:

- Validation performance

- Problem requirements

- Interpretability

- Generalization

- Dataset size

- Training speed

- Prediction speed